

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

YELAHANKA, BENGALURU – 560064



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Internship Report

2023-24 Seventh Semester

“Autism Spectrum Disorder Detection using Machine Learning”

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CERTIFICATE

This is to certify that the Internship titled “**Autism Spectrum Disorder Detection using Machine Learning**” carried out by Srikar G S S (1BY20IS166), Shreya R (1BY20IS158), Prajwal N (1BY20CS135), Pranav B C (1BY20CS224), bonafide students of BMS Institute of Technology and Management, in partial fulfillment for the award of Bachelor of Engineering, in Information Science and Computer Science and Engineering under Visvesvaraya Technological University, Belagavi, during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

Signature of Guide

Signature of HOD



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

DECLARATION

We, Srikar G S S (1BY20IS166), Shreya R (1BY20IS158), Prajwal N (1BY20CS135), Pranav B C (1BY20CS224), final year students of Bachelor of Engineering Artificial Intelligence and Machine Learning, BMS Institute of Technology and Management, Bengaluru, declare that the Internship has been successfully completed, in ZetaCoding Innovative Solutions. This report is submitted in partial fulfillment of the requirements for award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Place: Bengaluru

Date: 27-09-2023

**Srikar G S S
Shreya R
Prajwal N
Pranav B C**

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

INSTITUTE VISION

To emerge as one of the finest technical institutions of higher learning, to develop engineering professionals who are technically competent, ethical and environment friendly for betterment of the society.

INSTITUTE MISSION

Accomplish stimulating learning environment through high quality academic instruction, innovation and industry-institute interface.

DEPARTMENT VISION

To develop professionals equipped to build sustainable and intelligent solutions that effectively interact with the natural intelligence towards creating a digitally empowered environment for future generations, safeguarding social ethics.

DEPARTMENT MISSION

- ❖ To enable students with the spirit and power of interdisciplinary acumen by integrating a world of knowledge into a world of intelligent systems and subsystems.
- ❖ Boost academic outcome through place-based education and collaborations with established research labs and industries.
- ❖ Encourage entrepreneurship efforts among students and develop them into great leaders.

CONTENT

SL NO.	DESCRIPTION	PAGE NO.
CHAPTER 1	About the Company	1
CHAPTER 2	Introduction	3
CHAPTER 3	Requirement Specification	6
CHAPTER 4	Requirements and System Design	9
CHAPTER 5	System Design	11
CHAPTER 6	Implementation	12
CHAPTER 7	Results	29
CHAPTER 8	Conclusion	33
CHAPTER 9	References	34

ABSTRACT

It is always a complex procedure to diagnose autism spectrum disorder (ASD) because there is no particular medical test for autism, like a blood test, to make a diagnosis for the disorder. Autism spectrum disorder is defined by the disability and impairments of participating in social communication or the presence of restricted or repetitive behaviors, or both.

It is a behaviorally diagnosed condition. To make a diagnosis, doctors look at the child's developmental history and behavior. Apparently, most children do not attain a proper diagnosis for autism until it is too late. In some cases, parents are reluctant to accept that their child's mental growth is not developing along with the child's physical growth.

This lateness in diagnosis hinders a child's ability to get the help they need to keep developing. It is important to diagnose ASD as early as possible through monitoring, screening, and evaluating a child's development so that we can ensure proper care and support for an autistic child to help them reach their full potential.

So, we are developing a system that will have the ability to diagnose Autism and come up with a reliable and effective conclusion even without the help of a professional. We hope this system will be very helpful for those concerned parents who are worried about their child's growth and activities, at the same time it will be very useful for the professionals.

CHAPTER 1

About The Company

1.1 Introduction to ZetaCoding Innovative Solutions

ZetaCoding Innovative Solutions, headquartered in Bengaluru, Karnataka, India, is a well- established service-based company specializing in software development and training. The company has emerged as a leader in delivering high-quality software solutions across a wide spectrum of technologies, catering to diverse software applications. With a steadfast commitment to excellence, ZetaCoding adheres to industry standards, including ISO and IEEE, ensuring the delivery of top-notch projects and services.

1.2 Core Activities

ZetaCoding Innovative Solutions is engaged in several core activities:

1. **Software Development:** The company is renowned for its expertise in developing bespoke software projects and applications. These solutions consistently adhere to rigorous ISO and IEEE standards, showcasing the organization's dedication to quality.
2. **Academic Support:** ZetaCoding plays a crucial role in supporting students in their academic journeys, offering guidance on IEEE projects and comprehensive training in contemporary software technologies.
3. **Professional Team:** The organization boasts a team of highly qualified, experienced, and proficient individuals who are adept at delivering exceptional products within specified timelines. Their dedication to excellence is a hallmark of the company's ethos.
4. **End-to-End Solutions:** ZetaCoding provides comprehensive end-to-end solutions, encompassing Project Management, Analysis, Development, Deployment, and ongoing customer support, ensuring the fulfillment of client requirements.

5. **Long-Term Relationships:** Building enduring relationships with clients is a core value at ZetaCoding. The organization's commitment to delivering superior services fosters trust and fosters long-term collaborations.

6. **Diverse Team:** The ZetaCoding team represents a diverse blend of professionals, including highly experienced engineers, accomplished academicians, skilled programmers, and adept business development managers. This diverse talent pool equips the organization to offer holistic solutions tailored to the unique needs of its clients.

1.3 Mission and Vision

The mission of ZetaCoding Innovative Solutions revolves around three key principles:

1. **Cutting-Edge Technology Solutions:** To provide innovative and advanced technology solutions.
2. **Training and Skill Development:** To offer comprehensive training programs and foster skill development.
3. **Empowering Future Generations:** To enhance the productivity and capabilities of upcoming generations through education and technology.

ZetaCoding Innovative Solutions is dedicated to advancing technology, supporting education, and contributing to the growth and development of emerging professionals.

CHAPTER 2

INTRODUCTION

2.1 Autism Spectrum Disorder

Autism, formally known as Autism Spectrum Disorder (ASD), is a complex neurodevelopmental condition that affects individuals in various ways. It is characterized by a wide range of symptoms and challenges that primarily impact social interaction, communication, and behavior. Autism is often described as a "spectrum" disorder because it manifests differently in each person, with varying degrees of severity and a diverse array of strengths and difficulties.

Autism typically becomes apparent in early childhood, often before the age of three. However, it can continue to influence an individual's life throughout their lifespan. While the exact cause of autism remains the subject of ongoing research, it is believed to result from a combination of genetic and environmental factors.

Some common features of autism include:

Social Challenges: Individuals with autism may have difficulty with social interactions, including understanding social cues, maintaining eye contact, and forming relationships.

Communication Difficulties: Many people with autism experience challenges in verbal and nonverbal communication. Some may have delayed speech development, while others may struggle with expressive and receptive language skills.

Repetitive Behaviors: Individuals with autism often engage in repetitive behaviors or routines. These behaviors can include hand-flapping, rocking, or an intense focus on specific interests or topics.

Sensory Sensitivities: Sensory sensitivities are common in autism. Individuals may be hypersensitive (overreacting to sensory stimuli like lights, sounds, or textures) or hyposensitive (underreacting to sensory stimuli).

Strong Interests: Many individuals with autism have intense and focused interests in specific subjects or activities, often referred to as "special interests." These interests can be a source of passion and expertise.

Rigidity and Resistance to Change: Individuals with autism may prefer routines and can become distressed when faced with unexpected changes.

It's important to note that autism is a lifelong condition, and while there is no cure, early intervention and various therapies, such as speech therapy, occupational therapy, and behavioral interventions, can significantly improve the quality of life for individuals with autism. Additionally, with appropriate support and understanding from family, friends, and the community, many individuals with autism can lead fulfilling and successful lives, making unique contributions to society. Autism awareness and acceptance are crucial in promoting inclusivity and ensuring that individuals with autism are given the opportunities they deserve to reach their full potential.

2.2 Outline

According to the WHO's statistical data, 0.63% of children are diagnosed with Autistic Spectrum Disorder (ASD). ASD occurs in children and continues to expand to adolescents and adults. In most cases, symptoms appear within the first 5 years of life. ASD is a serious neurodevelopmental disorder with very high healthcare costs. ASD is characterized by a permanent lack of social interaction and stereotypical behavior, often accompanied by a widespread deterioration of communication skills

If ASD can be diagnosed early, it can be treated. Early detection of ASD plays a vital role in clinical diagnosis. This diagnosis can then inform personalized treatment regimens to improve the quality of life of children with ASD and their families. Unfortunately, the process of diagnosis of ASD can take a long time with expensive testing methods. Recently, the increase in ASD cases around the world is a motivation for doctors/scientists to find more effective screening methods.

Today, with the advancement in modern technology, we can store a huge amount of data. Data mining is an important task to make decisions based on the collected data. This is linked to machine learning. In recent years, machine learning has acquired many achievements and it is gaining more importance in applied sciences such as medicine and biomedicine. In clinical decision making and diagnostics, machine learning methods are used or recommended to support data interpretation. Therefore, methods of screening diseases with the aid of machine learning are extensively studied.

2.3 Problem Statement

To be able to detect Autism and be able to inform the user at an early stage of suspicion rather than wait for a longer period of time of detection process. Our aim is to be able to build a user friendly system that can be utilized to effectively detect the presence of Autism in the individual by collecting their responses to some questions.

As we have seen in the previous sections, detection at an early stage plays a vital role for any individual, so that an apt solution can be recommended by the doctor at the earliest to make the quality of life better.

2.4 Objectives

The main objective is to develop an application which can allow the user to be able to find out whether he/she may have chances of being Autistic. Other objectives of this application is to be able to satisfy the need for early detection of Autism and provide necessary details about such a medical condition. The application also consists of available doctors who specialize in certain fields, this can be vital and informative to the user. Our aim is to make this web application user friendly and accessible.

2.5 Project Application

Early Autism Diagnosis Tool: Develop an ML model that can analyze various data sources (speech patterns, eye movement, facial expressions, etc.) to detect early signs of Autism Spectrum Disorder (ASD) in children. This tool could be used in schools or healthcare settings.

Interactive system: Takes responses from the user to a set of predefined questions to analyse the symptoms or nature of the individual to be able to provide accurate prediction.

Doctor details: Application also consists of doctors details and information, this can be accessed by the user for their purposes. Helps to be able to identify necessary doctors based on condition and contact them.

Parental support: Detecting autism at an early stage and age is important so that the individual's life quality can be better or improved. So parents who suspect the condition of autism in their children can also use this application to be able to get an accurate prediction.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Overall System Description

The user is first provided with the website's index/home page. The user can access the ASD prediction form, doctors, about us, contact us and login/register pages.

Modules:

- ➔ **ASD predictor** : It will be redirected to a form that runs on streamlit python file where user fills the form and the response will be submitted to the machine learning algorithm.
- ➔ **Login/Register** : User will be redirected to a login register page where the user's name, email and other personal information will be registered onto the database when the user logs in for the first time.
- ➔ **Doctor** : Details of the doctors with their speciality of ASD treatment will be displayed for the users to choose the doctor they want to consult.
- ➔ **Book Appointment** : In this module, the user is required to enter the time and date for the appointment along with the doctor available. This data will be stored in the firebase database for the admin to monitor.
- ➔ **Admin** : This module is the server module, which receives data from the patients/ users and monitors redundant or duplicate records in the database and assigns doctors through booking appointments. Admin has access to all the other modules.

3.2 Hardware Requirements

- **Anaconda Navigator - Streamlit**
- **OS: Windows/ Linux**
- **Processor: Intel core i5 processor**
- **GPU: NVIDIA GeForce/A100**
- **Memory: 15.6 GB**
- **Hard Disk: 40 GB**

3.3 Software Requirements

Functional Requirements

User Registration and Authentication:

User registration and login functionality.

User profile creation and management.

Users can book appointments and can view the appointment details in their dashboard.

Machine Learning Models :

System should implement machine learning models suitable for autism detection.

Common models include logistic/linear regression, support vector machines, decision trees, KNN and more.

It should support training, validation, and evaluation of these models using appropriate metrics.

Real-time or Batch Processing:

Depending on the use case, the system should support both real-time(online) processing of data for autism detection

User Interface and dashboard:

There should be a user-friendly interface for clinicians to input patient data, initiate the autism detection process, and view the results.

User has their own profile where details of the booked appointment is displayed.

Cross-Validation and Hyperparameter Tuning:

The system should perform cross-validation and hyperparameter tuning to optimize the performance of the machine learning models and prevent overfitting.

Non-Functional Requirements

Availability

The online registration system shall permit backing up of the registration database while other registration activities are going on.

Online booking appointment is also provided for the patient to book an appointment with the required doctor if ASD predictor comes positive.

Accessibility

The system shall be accessible by people with specific vision needs to the extent that a user shall be able to display the whole user interface in a larger font without truncating displayed text or other values.

Security

The access permissions for system data may only be changed by the systems data administrator passwords shall never be viewable at the point of entry or any other time.

Database is only accessible to admin and no private information of the patient is gathered for security purposes.

CHAPTER 4

SYSTEM ANALYSIS

Machine learning models were prepared by executing multiple algorithms like Linear Regression, KNN, Random Forest and Decision Trees.

A class diagram was initially created to understand the dataflow of the modules integrating both the frontend and backend.

The training datasets are fetched from an existing dataset provided. The training of the model was performed in jupyter notebook as well as google-collab platforms.

Python language using pandas to implement the data sets into records and tables and implementing algorithms onto this training dataset to create models.

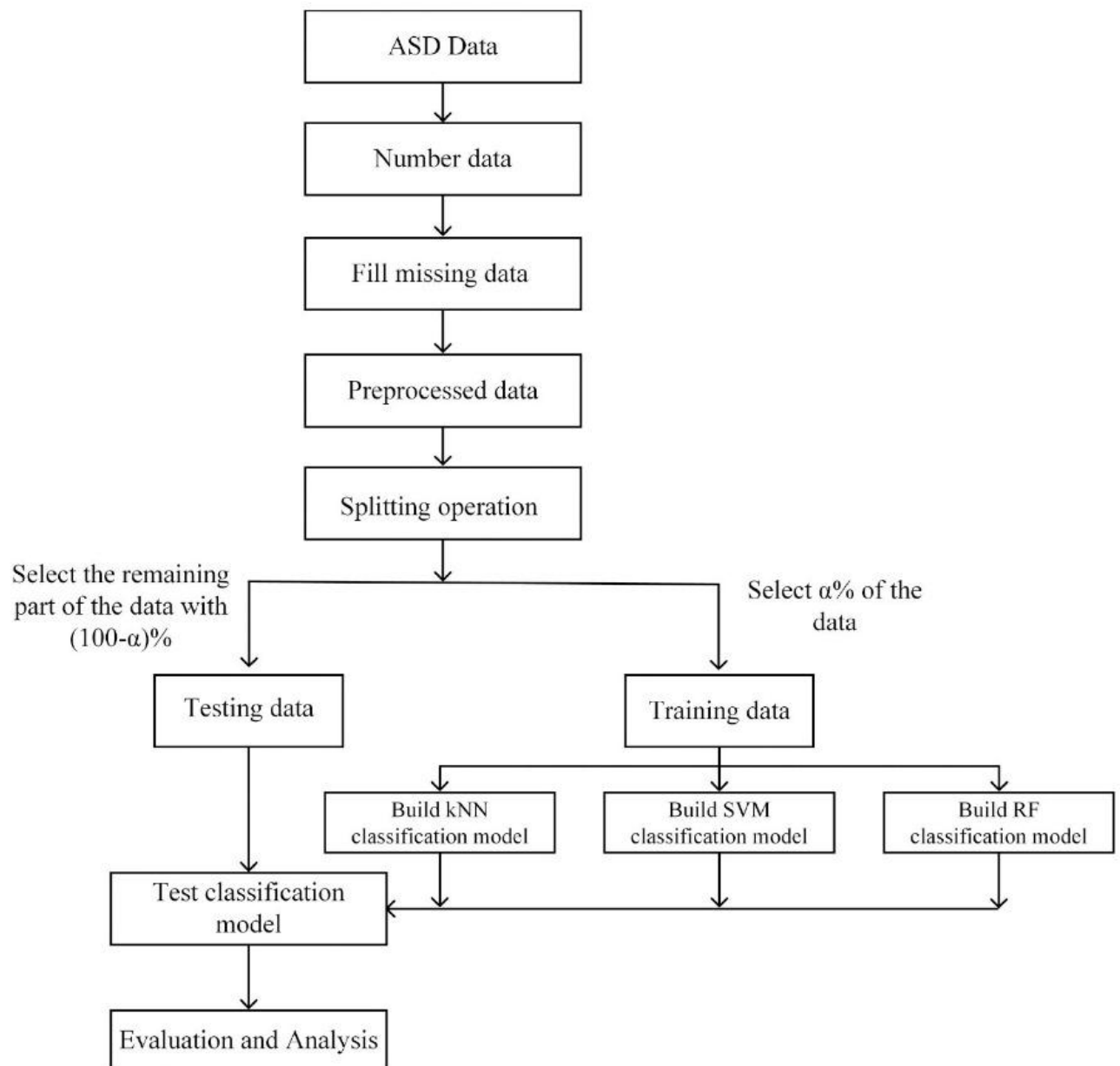
Once these models were created the algorithm which gave the best efficiency during the validation process was recorded.

Using a streamlit platform the form was created for the user to fill as an input, which will be converted into an array for the machine learning model to compare the input with the dataset and give out the output whether the person has chances of having ASD or not.

For frontend, a website with Homepage, Login and Registering page, Booking appointment, a userprofile page with dashboard, a gallery page is implemented using HTML, CSS and JavaScript languages.

The database is set in Firebase platform. A realtime database which updates and stores a new register of the patient as well as the patient's book appointment details. The cloud database is scalable and handles large amounts of data.

The whole project is built on Visual Studio code and runs on a local server/host which acts as both client and a server.



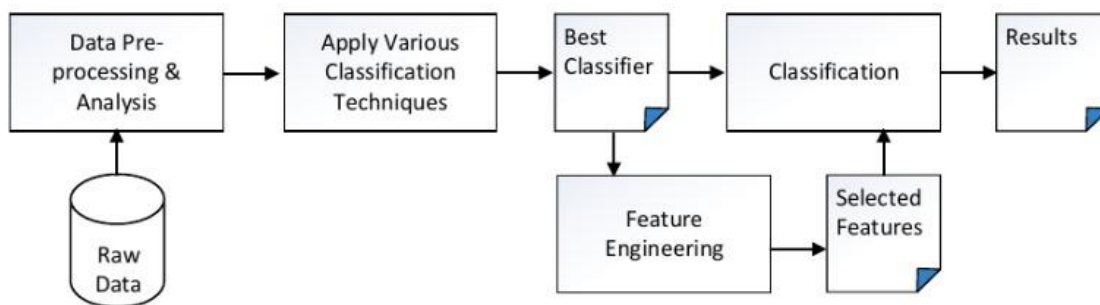
FLOWCHART OF ASD DETECTION

The graphs and statistics of each of the models were derived for analysis and in all the above cases with three ASD datasets, the kNN method gave the lowest scores. Whereas RF and SVM models gave the best efficient outputs.

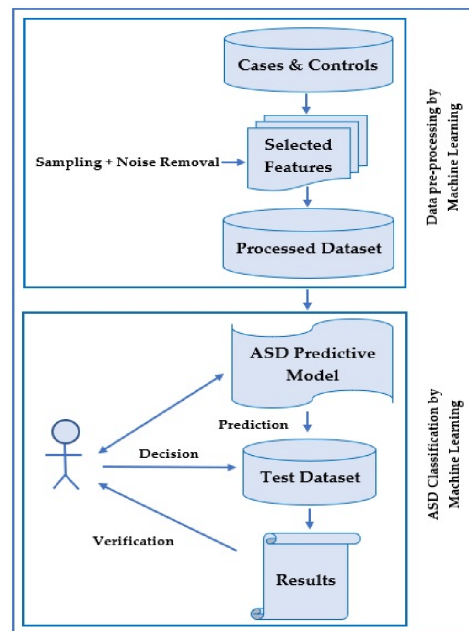
CHAPTER 5

SYSTEM DESIGN

Here's a simple diagram and class diagram of how the detection of ASD was analyzed and trained for implementation. As raw data is fetched for data preprocessing and later classified into clustering and regression of data.



METHODOLOGY OF DETECTION



CLASS DIAGRAM OF ASD DATASET TRAINING

CHAPTER 6

IMPLEMENTATION

3.1 EXPLANATION OF THE PROCEDURE FOLLOWED WITH CODE:

DATASET ANALYSIS AND PREPROCESSING:

Reading the dataset and creating a dataframe to work on:

```
In [3]: import pandas as pd
import seaborn as sns

In [4]: df=pd.read_csv('Toddler Autism dataset July 2018.csv')
```

```
In [6]: pd.read_csv('Toddler Autism dataset July 2018.csv')
```

```
Out[6]:
```

	Case_No	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Age_Mons	Qchat-10-Score	Sex	Ethnicity	Jaundice	Family_mem_with_ASD	Who completed the test	Class/ASD Traits	
0	1	0	0	0	0	0	0	1	1	0	1	28	3	f	middle eastern	yes		no	family member	No
1	2	1	1	0	0	0	1	1	0	0	0	36	4	m	White European	yes		no	family member	Yes
2	3	1	0	0	0	0	0	1	1	0	1	36	4	m	middle eastern	yes		no	family member	Yes
3	4	1	1	1	1	1	1	1	1	1	1	24	10	m	Hispanic	no		no	family member	Yes
4	5	1	1	0	1	1	1	1	1	1	1	20	9	f	White European	no		yes	family member	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1049	1050	0	0	0	0	0	0	0	0	0	1	24	1	f	White European	no		yes	family member	No
1050	1051	0	0	1	1	1	0	1	0	1	0	12	5	m	black	yes		no	family member	Yes
1051	1052	1	0	1	1	1	1	1	1	1	1	18	9	m	middle eastern	yes		no	family member	Yes
1052	1053	1	0	0	0	0	0	0	1	0	1	19	3	m	White European	no		yes	family member	No
1053	1054	1	1	0	0	1	1	0	1	1	0	24	6	m	Asian	yes		yes	family	Yes

Checking for null values in the rows and columns:

```
In [7]: df.isna().sum(axis=0) #empty fields in columns
```

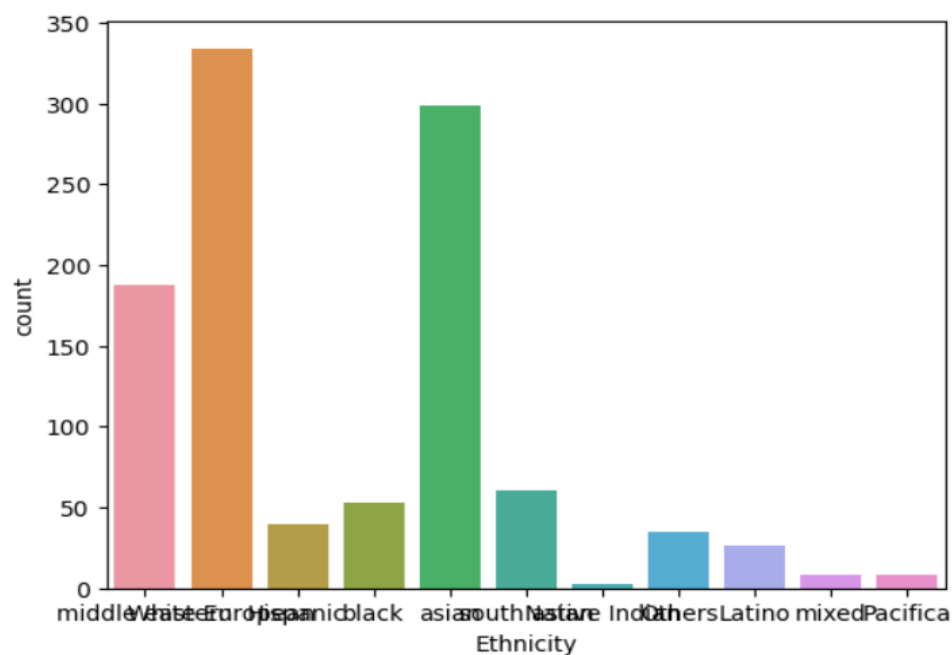
```
Out[7]: Case_No      0
A1      0
A2      0
A3      0
A4      0
A5      0
A6      0
A7      0
A8      0
A9      0
A10     0
Age_Mons 0
Qchat-10-Score 0
Sex      0
Ethnicity 0
Jaundice 0
Family_mem_with_ASD 0
Who completed the test 0
Class/ASD Traits 0
dtype: int64
```

```
In [8]: df.isna().sum(axis=1) #empty fields in rows
```

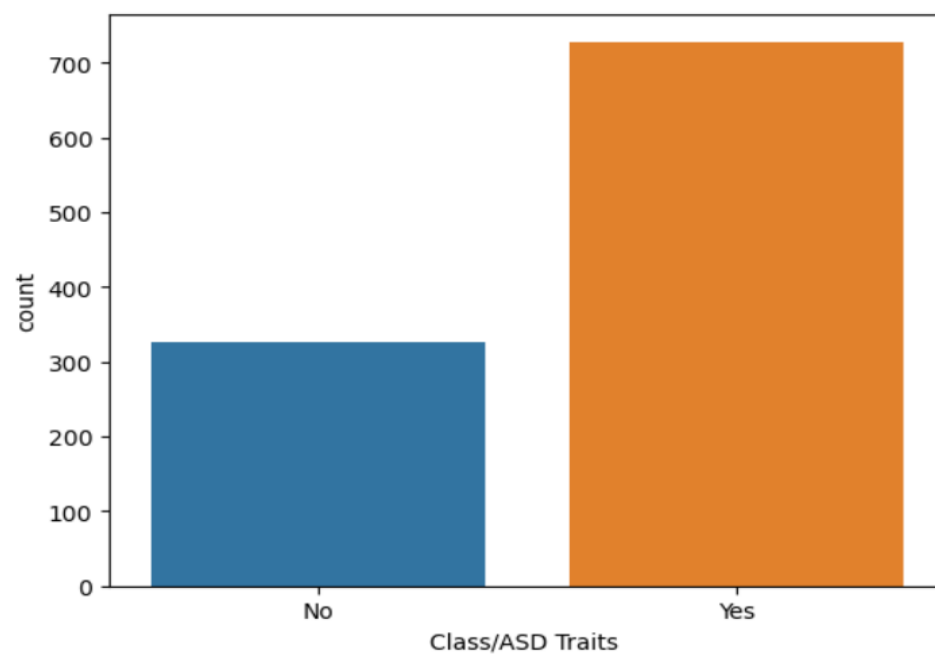
```
Out[8]: 0      0
1      0
2      0
3      0
4      0
...
1049    0
1050    0
1051    0
1052    0
```

Univariate and Bivariate analysis of all columns of the dataset:

```
Out[15]: <AxesSubplot:xlabel='Ethnicity', ylabel='count'>
```

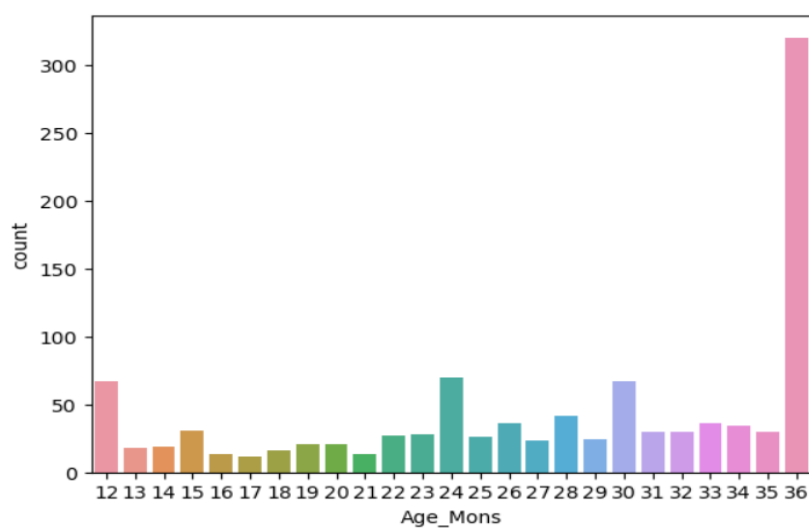


```
Out[17]: <AxesSubplot:xlabel='Class/ASD Traits ', ylabel='count'>
```



output column: Class/ASD Traits:

```
Out[20]: <AxesSubplot:xlabel='Age_Mons', ylabel='count'>
```

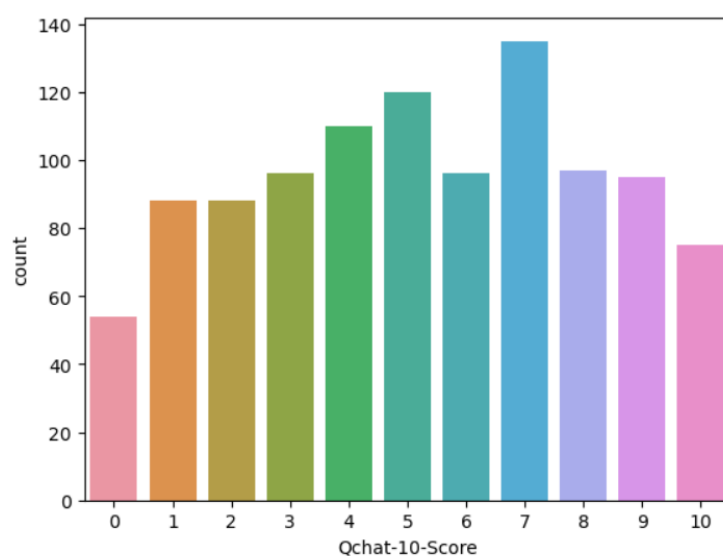


Age groups of Autism Individuals:

```
In [21]: df['Age_Mons'].value_counts()
```

```
Out[21]: 36    320
         24     70
         30     67
         12     67
         28     42
         26     36
         33     36
         34     34
         15     31
         31     30
         35     30
         32     30
         23     28
         22     27
         25     26
         29     24
         27     23
         19     21
         20     21
         14     19
         13     18
         18     16
         21     13
         16     13
         17     12
         Name: Age_Mons, dtype: int64
```

```
Out[23]: <AxesSubplot:xlabel='Qchat-10-Score', ylabel='count'>
```



```
In [24]: df['Qchat-10-Score'].value_counts()
```

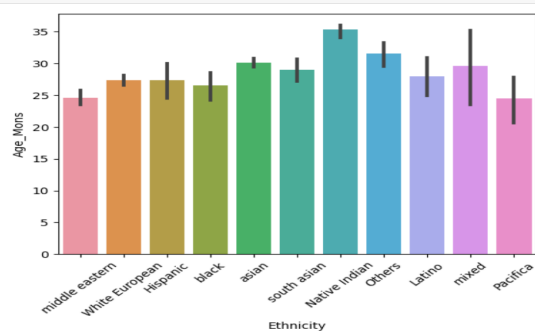
```
Out[24]: 7      135
         5      120
         4      110
         8       97
         3       96
         6       96
         9       95
         2       88
         1       88
        10       75
         0       54
         Name: Qchat-10-Score, dtype: int64
```

```
In [25]: df['Sex'].value_counts()
```

```
Out[25]: m      735
         f      319
         Name: Sex, dtype: int64
```

Analyzing the Sex of Autistic Individuals-counts

```
In [6]: sns.barplot(df['Ethnicity'], df['Age_Mons'])
         plt.xticks(rotation=45)
         plt.show()
```



```
In [37]: df['who completed the test'].value_counts()
```

```
Out[37]: family member      1018
Health Care Professional    24
Health care professional     5
Self                        4
Others                      3
Name: who completed the test, dtype: int64
```

```
In [38]: df['who completed the test']=df['who completed the test'].str.lower()
```

```
In [39]: df
df['who completed the test'].value_counts()
```

```
Out[39]: family member      1018
health care professional    29
self                       4
others                     3
Name: who completed the test, dtype: int64
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Age_Mons
count	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1054.000000	1
mean	0.563567	0.448767	0.401328	0.512334	0.524668	0.576850	0.649905	0.459203	0.489564	0.586338	27.867173
std	0.496178	0.497604	0.490400	0.500085	0.499628	0.494293	0.477226	0.498569	0.500128	0.492723	7.980354
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	12.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	23.000000
50%	1.000000	0.000000	0.000000	1.000000	1.000000	1.000000	1.000000	0.000000	0.000000	1.000000	30.000000
75%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	36.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	36.000000

Train Test Split:

1054 rows x 11 columns

```
In [28]: #Preprocessing Done

from sklearn.model_selection import train_test_split
from sklearn.feature_selection import RFECV
from sklearn.model_selection import GridSearchCV , KFold , cross_val_score, ShuffleSplit, cross_validate
```

```
In [29]: Y=df['Class/ASD Traits ']
X=df.drop(['Class/ASD Traits '],axis=1)
x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.20,random_state=7)
print('Processed....')

Processed....
```

Bucketing was done to Age_Mons column to simplify the process:

```
In [1]: age_bins = [0, 12, 18, 24, 40] # You can define your own bin edges

# Define bin labels
age_labels = ['12-17', '18-23', '24-29', '30-36']

# Add a new column with bucket labels
df['Age_Mons'] = pd.cut(df['Age_Mons'], bins=age_bins, labels=age_labels, right=False)
print(df)
```

X_Train and X_Test data:

```
In [53]: x_train
```

```
Out[53]:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Age_Mons	Qchat-10-Score	Sex	Ethnicity	Jaundice	Family_mem_with_ASD
779	1	1	1	1	1	1	1	1	1	0	2	9	1	6	0	0
328	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0
649	0	0	1	1	0	0	0	1	0	1	2	4	1	6	0	0
137	0	0	1	1	1	0	0	0	1	0	2	4	1	6	1	0
1011	0	1	0	0	1	1	0	0	1	0	2	4	1	5	0	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
919	0	0	0	0	1	0	1	0	0	1	2	3	1	10	0	0
211	1	1	0	0	1	1	1	1	1	0	2	7	1	6	0	0
502	0	1	1	1	1	1	1	0	1	1	0	8	1	5	0	1
537	1	1	0	1	1	1	0	0	0	0	2	5	1	3	0	0
175	1	0	0	0	1	1	0	1	0	1	2	5	1	6	0	1

843 rows x 16 columns

```
In [54]: x_test
```

```
Out[54]:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Age_Mons	Qchat-10-Score	Sex	Ethnicity	Jaundice	Family_mem_with_ASD
650	0	0	1	1	0	1	0	0	0	1	2	4	1	6	0	0
34	0	0	0	0	0	0	0	0	0	1	0	1	0	8	0	0
54	0	0	0	0	0	0	0	0	0	1	0	1	0	7	0	0
656	1	1	1	1	1	1	1	0	1	0	1	8	1	8	1	0
874	0	0	1	1	0	0	0	1	1	0	2	4	1	10	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
673	1	0	1	1	1	1	1	1	1	1	2	9	0	6	0	0
662	1	0	0	1	1	1	1	0	0	0	2	5	0	5	0	0
699	1	1	0	0	0	1	1	1	1	1	0	7	0	5	1	0
534	1	1	1	1	1	1	1	1	1	0	0	9	1	0	1	0
990	1	0	1	1	1	1	1	0	1	1	2	8	0	8	0	0

```
In [23]: y_train
```

```
Out[23]:
```

779	1
328	0
649	1
137	1
1011	1
...	..
919	0
211	1
502	1
537	1
175	1

Name: Class/ASD Traits , Length: 843, dtype: int64

```
In [24]: y_test
```

```
Out[24]:
```

650	1
34	0
54	0
656	1
874	1
...	..
673	1
662	1
699	1
534	1
990	1

Name: Class/ASD Traits , Length: 211, dtype: int64

Y_Train and Y_Test data:

```
In [23]: y_train
Out[23]: 779    1
          328    0
          649    1
          137    1
          1011   1
          ..
          919    0
          211    1
          502    1
          537    1
          175    1
          Name: Class/ASD Traits , Length: 843, dtype: int64
```

```
In [24]: y_test
Out[24]: 650    1
          34    0
          54    0
          656    1
          874    1
          ..
          673    1
          662    1
          699    1
          534    1
          990    1
          Name: Class/ASD Traits , Length: 211, dtype: int64
```

Encoding of the dataframe:

We used OneHotEncoder to perform encoding operations.

```
In [7]: from sklearn.preprocessing import OneHotEncoder

In [8]: #Encoding from edf to
encoder=OneHotEncoder(handle_unknown='ignore')

In [9]: encoder_df=pd.DataFrame(encoder.fit_transform(edf[['Ethnicity']]).toarray())

In [10]: encoder_df
Out[10]:
```

	0	1	2	3	4	5	6	7	8	9	10
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
3	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...
1049	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
1050	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0
1051	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
1052	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
1053	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0

1054 rows × 11 columns

	Hispanic	Latino	Native Indian	Others	Pacafica	White European	\
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	1.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	1.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	1.0	0.0
...	...	...	...	...	...	...	...
1049	0.0	0.0	0.0	0.0	0.0	1.0	0.0
1050	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1051	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1052	0.0	0.0	0.0	0.0	0.0	1.0	0.0
1053	0.0	0.0	0.0	0.0	0.0	0.0	0.0

	Asian	Black	Middle Eastern	Mixed	South Asian
0	0.0	0.0	1.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	1.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...
1049	0.0	0.0	0.0	0.0	0.0
1050	0.0	1.0	0.0	0.0	0.0
1051	0.0	0.0	1.0	0.0	0.0
1052	0.0	0.0	0.0	0.0	0.0
1053	1.0	0.0	0.0	0.0	0.0

[1054 rows x 11 columns]

All Categorical columns encoded

Importing all necessary libraries to implement all Algorithms on the processed Data:

```
In [40]: # Importing libraries
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.impute import SimpleImputer
# Suppress warnings
import warnings
warnings.filterwarnings("ignore")

# Classification
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.svm import SVC, LinearSVC
from sklearn.ensemble import RandomForestClassifier

# Regression
from sklearn.linear_model import LinearRegression, Ridge, Lasso, RidgeCV, ElasticNet, LogisticRegression
from sklearn.ensemble import RandomForestRegressor, BaggingRegressor, GradientBoostingRegressor, AdaBoostRegressor

# Modelling Helpers :
from sklearn.preprocessing import Normalizer, scale
from sklearn.model_selection import train_test_split
from sklearn.feature_selection import RFECV
from sklearn.model_selection import GridSearchCV, KFold, cross_val_score, ShuffleSplit, cross_validate

# Preprocessing :
from sklearn.preprocessing import MinMaxScaler, StandardScaler, LabelEncoder

# Metrics :
# Regression
from sklearn.metrics import mean_squared_log_error, mean_squared_error, r2_score, mean_absolute_error
# Classification
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report

print("Setup complete...")

Setup complete...
```

LOGISTIC REGRESSION ALGORITHM:

```
In [5]: #LOGISTIC REGRESSION
```

```
In [6]: from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix
logreg = LogisticRegression()
```

```
In [7]: logreg.fit(x_train, y_train)
```

```
Out[7]: LogisticRegression()
```

```
In [8]: preds = logreg.predict(x_test)
```

```
In [9]: logreg.score(x_train, y_train)
```

```
Out[9]: 1.0
```

```
In [10]: classification_report(y_test, preds)
```

```
Out[10]: '
           precision    recall  f1-score   support\n\n
  0       1.00        1.00        1.00         95\n
  1       1.00        1.00        1.00        222\n\n
 accuracy                   1.00         317\n
 macro avg                 1.00         317\n
 weighted avg              1.00         317\n'
```

```
In [11]: from sklearn.metrics import classification_report, confusion_matrix
CRMLR=classification_report(y_test, preds)#CRMLR is ClasificationReport Matrix Var
print(CRMLR)
```

```

           precision    recall  f1-score   support

    0       1.00        1.00        1.00         95
    1       1.00        1.00        1.00        222

 accuracy                   1.00         317
 macro avg                 1.00         317
 weighted avg              1.00         317
```

DECISION TREE ALGORITHM:

```
In [13]: #DecisionTree
```

```
In [14]: from sklearn.tree import DecisionTreeClassifier  
dtree = DecisionTreeClassifier()  
dtree.fit(x_train, y_train)  
pred = dtree.predict(x_test)  
classification_report(y_test,pred)
```

```
Out[14]: '          precision    recall  f1-score   support\n\n     0       0.85      0.85      0.85         95\n     1       0.94      0.94      0.94        222\n\n accuracy               0.91      317\n macro avg              0.89      317\nweighted avg              0.91      317'
```

```
In [15]: CMDT=confusion_matrix(y_test,pred)  
print(CMDT)
```

```
[[ 81 14]  
 [ 14 208]]
```

```
In [16]: CRDT=classification_report(y_test,pred)  
print(CRDT)
```

```
          precision    recall  f1-score   support\n\n     0       0.85      0.85      0.85         95\n     1       0.94      0.94      0.94        222\n\n accuracy               0.91      317\n macro avg              0.89      317\nweighted avg              0.91      317
```

RANDOM FOREST ALGORITHM:

```
In [17]: #RandomForest
```

```
In [18]: from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier()
rf.fit(x_train, y_train)
pred = rf.predict(x_test)
classification_report(y_test,pred)
```

```
Out[18]:
```

		precision	recall	f1-score	support							
1	0.95	0.98	0.96	0.96	222	accuracy	0.95	0.94	0.88	0.91	95	
0.93	0.94	317	weighted avg	0.95	0.95	0.95	317		0.95	317	macro avg	0.95

```
In [19]: CMRF=confusion_matrix(y_test,pred)
print(CMRF)
```

```
[[ 84 11]
 [ 5 217]]
```

```
In [20]: CRRF=classification_report(y_test,pred)
print(CRRF)
```

	precision	recall	f1-score	support
0	0.94	0.88	0.91	95
1	0.95	0.98	0.96	222
accuracy			0.95	317
macro avg	0.95	0.93	0.94	317
weighted avg	0.95	0.95	0.95	317

SVM [SUPPORT VECTOR MACHINE] ALGORITHM:

```
In [22]: #SVM
```

```
In [23]: from sklearn.svm import SVC
model = SVC()
model.fit(x_train, y_train)
pred = model.predict(x_test)
classification_report(y_test,pred)
```

```
Out[23]:
```

		precision	recall	f1-score	support							
1	0.98	0.99	0.98	0.98	222	accuracy	0.98	0.97	0.96	0.96	95	
0.97	0.97	317	weighted avg	0.98	0.98	0.98	317		0.98	317	macro avg	0.98

```
In [24]: CMSVM=confusion_matrix(y_test,pred)
print(CMSVM)
```

```
[[ 91  4]
 [ 3 219]]
```

```
In [25]: CRSVM=classification_report(y_test,pred)
print(CRSVM)
```

	precision	recall	f1-score	support
0	0.97	0.96	0.96	95
1	0.98	0.99	0.98	222
accuracy			0.98	317
macro avg	0.98	0.97	0.97	317
weighted avg	0.98	0.98	0.98	317

KNN ALGORITHM:

```

In [27]: #KNN

In [28]: knn = KNeighborsClassifier(n_neighbors=1)

In [29]: knn.fit(x_train, y_train)
Out[29]: KNeighborsClassifier(n_neighbors=1)

In [30]: pred = knn.predict(x_test)

In [31]: CRKNN=classification_report(y_test, pred)
print(CRKNN)

```

	precision	recall	f1-score	support
0	0.85	0.94	0.89	95
1	0.97	0.93	0.95	222
accuracy			0.93	317
macro avg	0.91	0.93	0.92	317
weighted avg	0.93	0.93	0.93	317

```

In [32]: error_rate = []
for i in range(1,40):
    knn = KNeighborsClassifier(n_neighbors=i)
    knn.fit(x_train, y_train)
    pred = knn.predict(x_test)
    error_rate.append(np.mean(y_test != pred))

```

OBSERVATIONS AFTER APPLYING ALL ALGORITHMS:

- On analyzing the accuracies of all algorithms, we opted SVM Algorithm to implement our project.

ACCURACIES:

Logistic Regression: 1.0

KNN: 0.9305993690851735

Decision Tree: 0.9053627760252366

Random Forest: 0.9589905362776026

SVM: 0.9779179810725552**CREATING A PICKLE FILE :**

```

29 # Corrected pickle.dump and pickle.load
30 with open('model.pkl', 'wb') as model_file:
31     pickle.dump(msvc, model_file)
32
33 # Loading the model
34 with open('model.pkl', 'rb') as model_file:
35     loaded_model = pickle.load(model_file)
36
37 model=joblib.load('model.pkl')

```

BUILDING THE WEB APPLICATION:

form page:

```

33     result[13:15] = [0, 1]
34
35     # Fill in jaundice-related elements
36     if jaundice == "Yes":
37         result[15:17] = [0, 1]
38     else:
39         result[15:17] = [1, 0]
40
41     # Fill in family autism-related elements
42     if family_autism == "Yes":
43         result[17:19] = [0, 1]
44     else:
45         result[17:19] = [1, 0]
46
47     # Define ethnicity categories
48     ethnicity_categories = {
49         "Hispanic": [1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
50         "Latino": [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
51         "Native Indian": [0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
52         "Others": [0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
53         "Pacific": [0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
54         "White European": [0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
55         "Asian": [0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
56         "Black": [0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
57         "Middle Eastern": [0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
58         "Mixed": [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0],
59         "South Asian": [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
60     }
61
62     # Fill in ethnicity-related elements
63     result[19:] = ethnicity_categories.get(ethnicity, [0] * 11)
64     print(result)

```

```

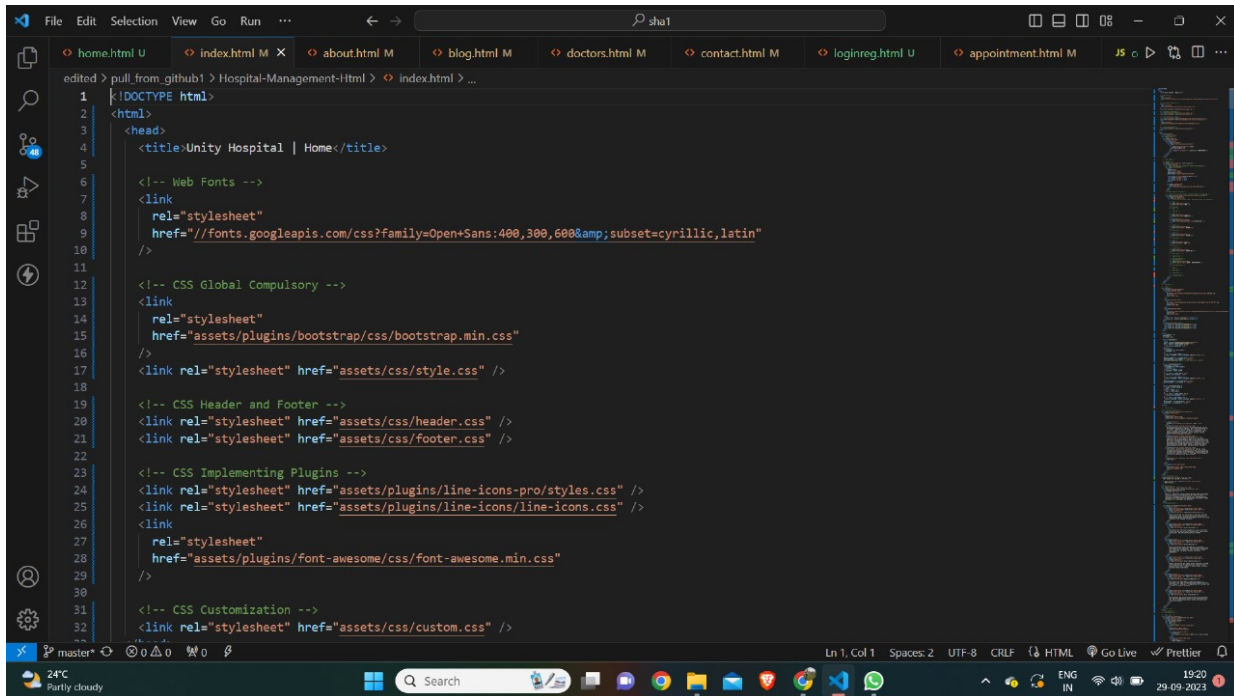
68     st.title("Autism Prediction App")
69
70     # Create a form to collect user inputs
71     with st.form("user_input_form"):
72         st.header("Please answer the following questions:")
73
74         # 10 questions with yes/no answers
75         questions = [
76             "I often notice small sounds when others do not:",
77             "I usually concentrate on whole picture rather than small details:",
78             "I find it easy to do more than one thing at once: ",
79             "If there is an interruption i can switch back to what i was doing very quickly:",
80             "I find it easy to read between the lines when someone is talking to me:",
81             "I know how to tell if someone listening to me is getting bored:",
82             "When am reading a story, i find it difficult to workout the characters intensions:",
83             "I like to collect information about categories of things:",
84             "I find it easy to workout what someone is thinking or feeling just by looking at their face:",
85             "I find it difficult to workout peoples intentions:"
86         ]
87         answers = [st.radio(q, ["Yes", "No"]) for q in questions]
88
89         st.header("Please provide additional information:")
90
91         # Age category
92         age = st.selectbox("Age", ["0-18 months", "19-30 months", "31-60 months"])
93
94         # Gender
95         gender = st.radio("Gender", ["Female", "Male"])
96
97         # Jaundice
98         jaundice = st.radio("Was the child born with jaundice?", ["Yes", "No"])

```

```
103 # Ethnicity
104 ethnicity = st.selectbox(
105     "Ethnicity",
106     [
107         "Hispanic",
108         "Latino",
109         "Native Indian",
110         "Others",
111         "Pacific",
112         "White European",
113         "Asian",
114         "Black",
115         "Middle Eastern",
116         "Mixed",
117         "South Asian",
118     ]
119 )
120
121 # Submit button
122 submit_button = st.form_submit_button(label="Submit")
123
124 # When the form is submitted
125 if submit_button:
126     # Generate the array based on user inputs
127     user_array = generate_array(answers, age, gender, jaundice, family_autism, ethnicity)
128
129     # Perform prediction using the loaded model (replace this with your actual prediction code)
130     prediction = model.predict([user_array])
131
132     # Display the prediction result
133     st.subheader("Prediction Result:")
134     #st.write("Autism Prediction:", prediction[0])
```

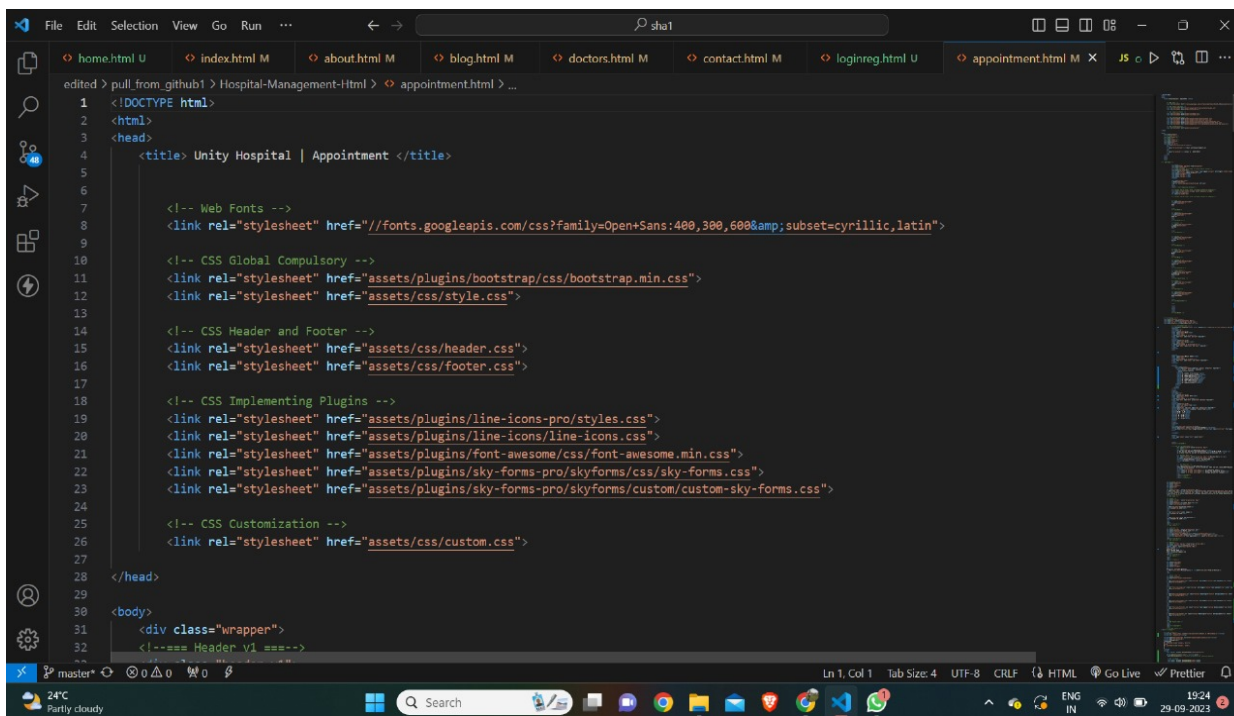
```
124 # When the form is submitted
125 if submit_button:
126     # Generate the array based on user inputs
127     user_array = generate_array(answers, age, gender, jaundice, family_autism, ethnicity)
128
129     # Perform prediction using the loaded model (replace this with your actual prediction code)
130     prediction = model.predict([user_array])
131
132     # Display the prediction result
133     st.subheader("Prediction Result:")
134     #st.write("Autism Prediction:", prediction[0])
135     if prediction == 1:
136         st.write("No autism detected.")
137     else:
138         st.write("Autism detected.")
139         st.write("Please visit [this link](https://www.google.com) for more information.")
140
141     print(user_array)
142
```

Main/Index page:



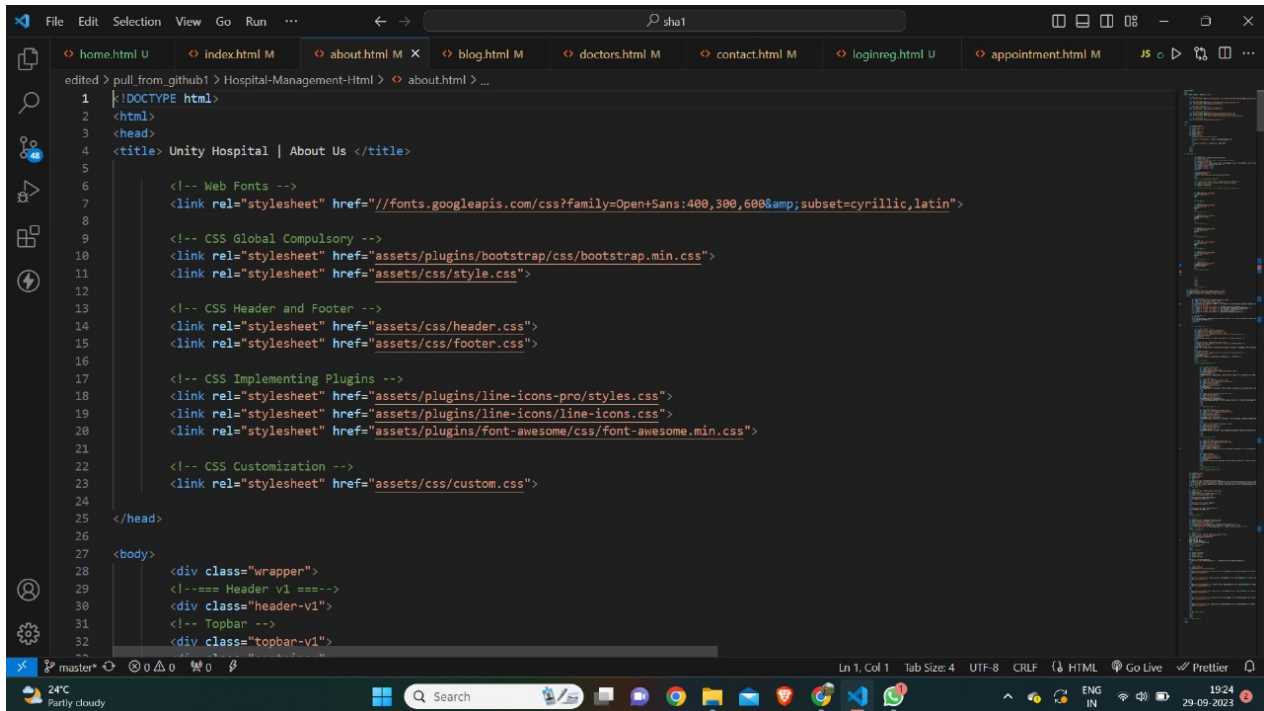
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Unity Hospital | Home</title>
5
6 <!-- Web Fonts -->
7 <link
8   rel="stylesheet"
9   href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin"
10  />
11
12 <!-- CSS Global Compulsory -->
13 <link
14   rel="stylesheet"
15   href="assets/plugins/bootstrap/css/bootstrap.min.css"
16  />
17 <link rel="stylesheet" href="assets/css/style.css" />
18
19 <!-- CSS Header and Footer -->
20 <link rel="stylesheet" href="assets/css/header.css" />
21 <link rel="stylesheet" href="assets/css/footer.css" />
22
23 <!-- CSS Implementing Plugins -->
24 <link rel="stylesheet" href="assets/plugins/line-icons-pro/styles.css" />
25 <link rel="stylesheet" href="assets/plugins/line-icons/line-icons.css" />
26 <link
27   rel="stylesheet"
28   href="assets/plugins/font-awesome/css/font-awesome.min.css"
29  />
30
31 <!-- CSS Customization -->
32 <link rel="stylesheet" href="assets/css/custom.css" />
```

Appointment Page:



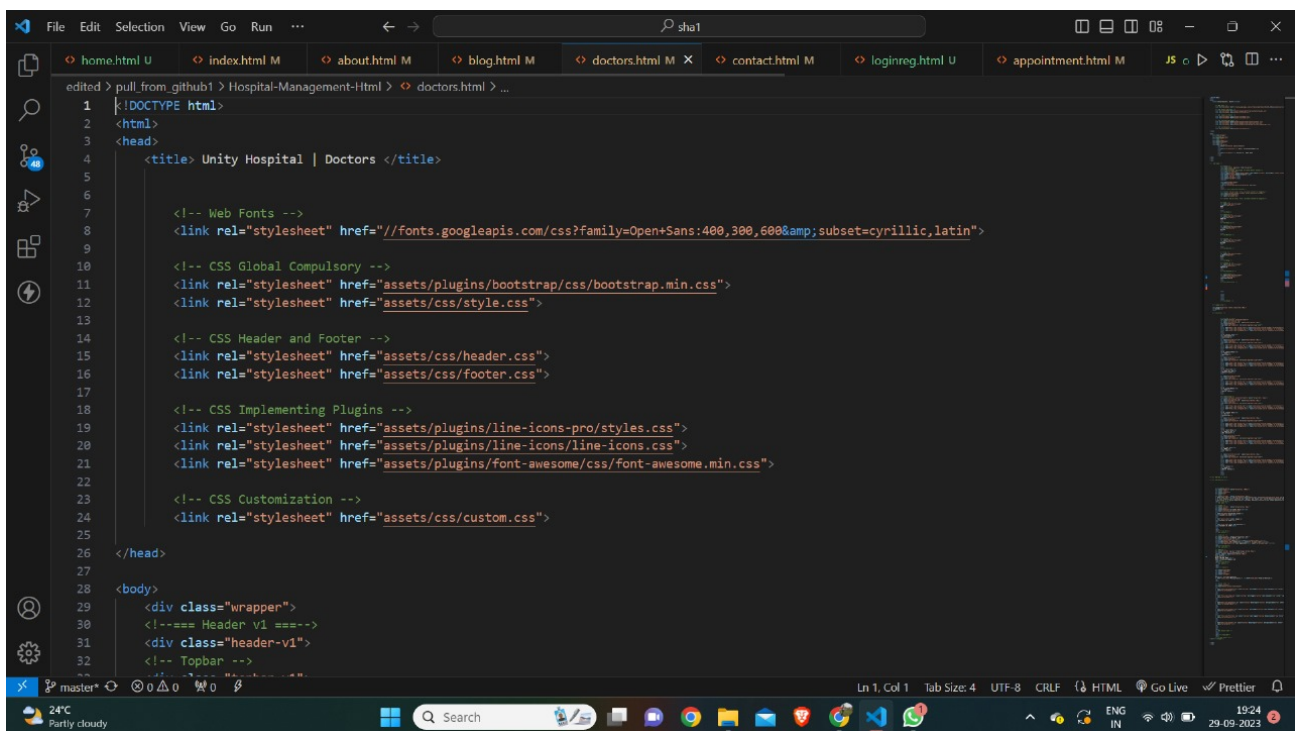
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Unity Hospital | Appointment </title>
5
6 <!-- Web Fonts -->
7 <link rel="stylesheet" href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin">
8
9 <!-- CSS Global Compulsory -->
10 <link rel="stylesheet" href="assets/plugins/bootstrap/css/bootstrap.min.css">
11 <link rel="stylesheet" href="assets/css/style.css">
12
13 <!-- CSS Header and Footer -->
14 <link rel="stylesheet" href="assets/css/header.css">
15 <link rel="stylesheet" href="assets/css/footer.css">
16
17 <!-- CSS Implementing Plugins -->
18 <link rel="stylesheet" href="assets/plugins/line-icons-pro/styles.css">
19 <link rel="stylesheet" href="assets/plugins/line-icons/line-icons.css">
20 <link rel="stylesheet" href="assets/plugins/font-awesome/css/font-awesome.min.css">
21 <link rel="stylesheet" href="assets/plugins/sky-forms-pro/skyforms/css/sky-forms.css">
22 <link rel="stylesheet" href="assets/plugins/sky-forms-pro/skyforms/custom/custom-sky-forms.css">
23
24 <!-- CSS Customization -->
25 <link rel="stylesheet" href="assets/css/custom.css">
26
27 </head>
28
29 <body>
30 <div class="wrapper">
31 <!--==== Header v1 =====>
```

About page:



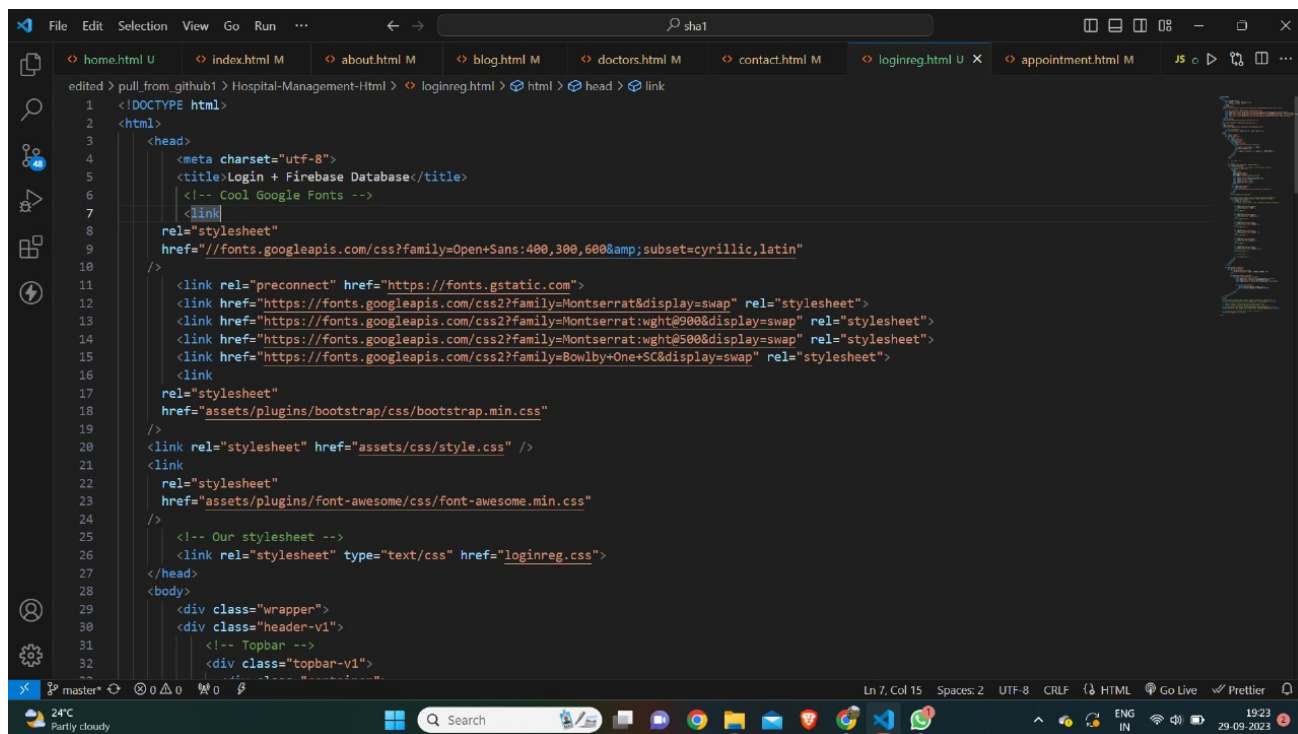
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title> Unity Hospital | About Us </title>
5
6 <!-- Web Fonts -->
7 <link rel="stylesheet" href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin">
8
9 <!-- CSS Global Compulsory -->
10 <link rel="stylesheet" href="assets/plugins/bootstrap/css/bootstrap.min.css">
11 <link rel="stylesheet" href="assets/css/style.css">
12
13 <!-- CSS Header and Footer -->
14 <link rel="stylesheet" href="assets/css/header.css">
15 <link rel="stylesheet" href="assets/css/footer.css">
16
17 <!-- CSS Implementing Plugins -->
18 <link rel="stylesheet" href="assets/plugins/line-icons-pro/styles.css">
19 <link rel="stylesheet" href="assets/plugins/line-icons/line-icons.css">
20 <link rel="stylesheet" href="assets/plugins/font-awesome/css/font-awesome.min.css">
21
22 <!-- CSS Customization -->
23 <link rel="stylesheet" href="assets/css/custom.css">
24 </head>
25
26 <body>
27 <div class="wrapper">
28 <!--==== Header v1 ====-->
29 <div class="header-v1">
30 <!-- Topbar -->
31 <div class="topbar-v1">
```

Doctors page:



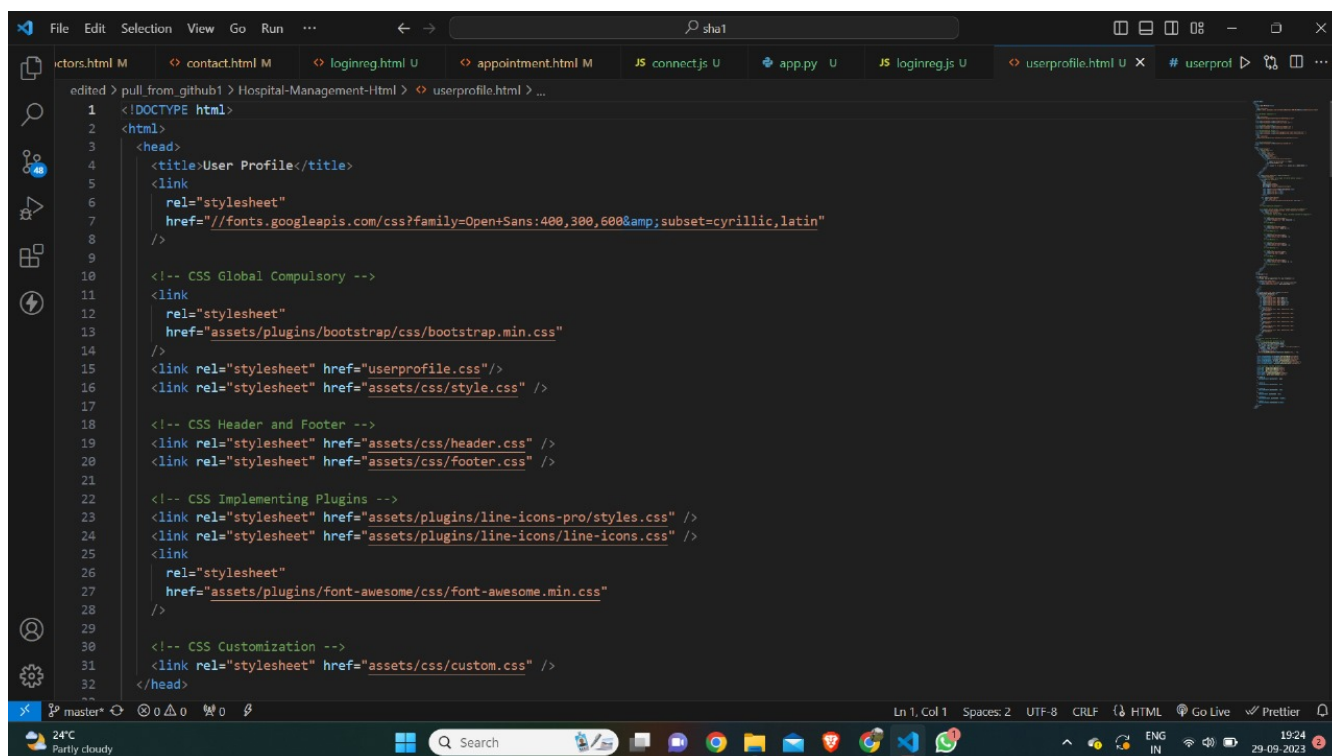
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title> Unity Hospital | Doctors </title>
5
6 <!-- Web Fonts -->
7 <link rel="stylesheet" href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin">
8
9 <!-- CSS Global Compulsory -->
10 <link rel="stylesheet" href="assets/plugins/bootstrap/css/bootstrap.min.css">
11 <link rel="stylesheet" href="assets/css/style.css">
12
13 <!-- CSS Header and Footer -->
14 <link rel="stylesheet" href="assets/css/header.css">
15 <link rel="stylesheet" href="assets/css/footer.css">
16
17 <!-- CSS Implementing Plugins -->
18 <link rel="stylesheet" href="assets/plugins/line-icons-pro/styles.css">
19 <link rel="stylesheet" href="assets/plugins/line-icons/line-icons.css">
20 <link rel="stylesheet" href="assets/plugins/font-awesome/css/font-awesome.min.css">
21
22 <!-- CSS Customization -->
23 <link rel="stylesheet" href="assets/css/custom.css">
24 </head>
25
26 <body>
27 <div class="wrapper">
28 <!--==== Header v1 ====-->
29 <div class="header-v1">
30 <!-- Topbar -->
31 <div class="topbar-v1">
```

Loginin page:



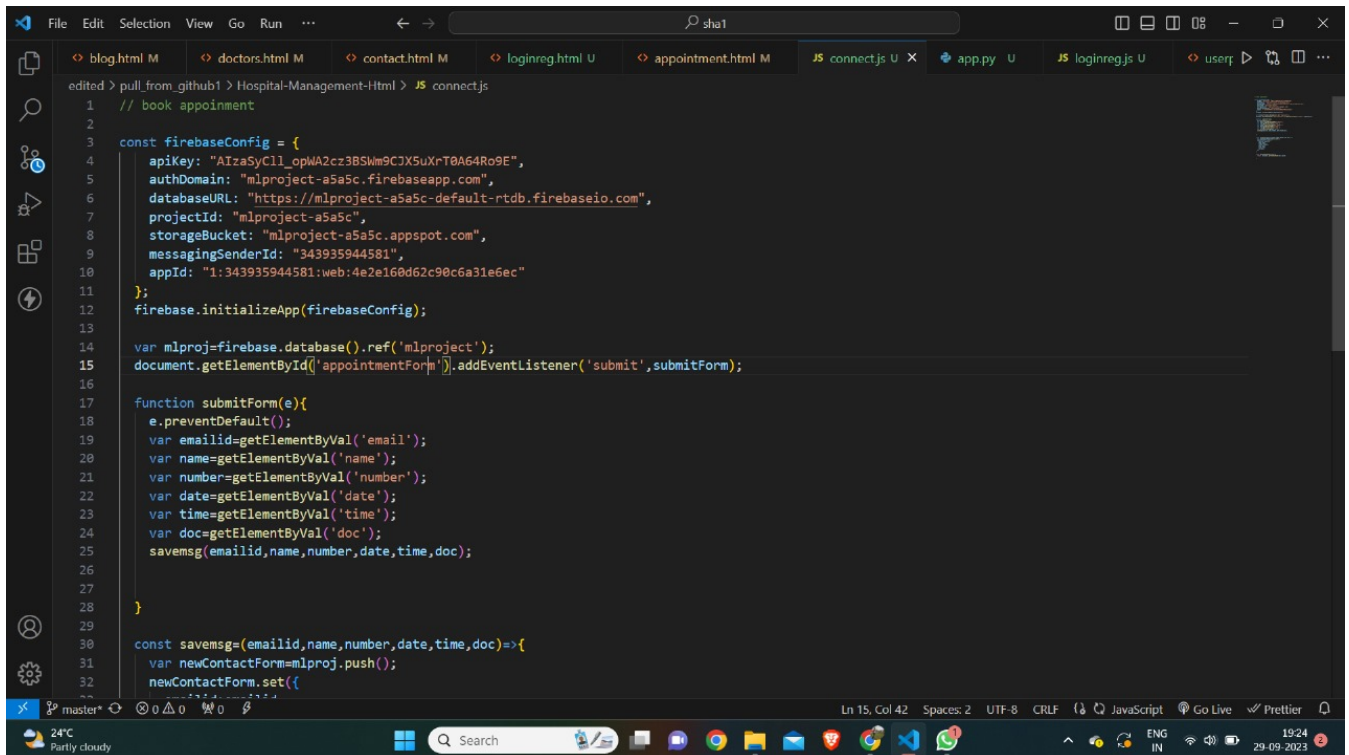
```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Login + Firebase Database</title>
6     <!-- Cool Google Fonts -->
7     <link
8       rel="stylesheet"
9       href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin"
10    />
11
12    <link rel="preconnect" href="https://fonts.gstatic.com">
13    <link href="https://fonts.googleapis.com/css2?family=Montserrat&display=swap" rel="stylesheet">
14    <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@900&display=swap" rel="stylesheet">
15    <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@500&display=swap" rel="stylesheet">
16    <link href="https://fonts.googleapis.com/css2?family=Bowlby+One+SC&display=swap" rel="stylesheet">
17
18    <link
19      rel="stylesheet"
20      href="assets/plugins/bootstrap/css/bootstrap.min.css"
21    />
22
23    <link rel="stylesheet" href="assets/css/style.css" />
24
25    <link
26      rel="stylesheet"
27      href="assets/plugins/font-awesome/css/font-awesome.min.css"
28    />
29
30    <!-- Our stylesheet -->
31    <link rel="stylesheet" type="text/css" href="loginreg.css">
32  </head>
33  <body>
34    <div class="wrapper">
35      <div class="header-v1">
36        <!-- Topbar -->
37        <div class="topbar-v1">
```

profile page:



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>User Profile</title>
5     <link
6       rel="stylesheet"
7       href="//fonts.googleapis.com/css?family=Open+Sans:400,300,600&subset=cyrillic,latin"
8     />
9
10    <!-- CSS Global Compulsory -->
11    <link
12      rel="stylesheet"
13      href="assets/plugins/bootstrap/css/bootstrap.min.css"
14    />
15
16    <link rel="stylesheet" href="userprofile.css"/>
17    <link rel="stylesheet" href="assets/css/style.css" />
18
19    <!-- CSS Header and Footer -->
20    <link rel="stylesheet" href="assets/css/header.css" />
21    <link rel="stylesheet" href="assets/css/footer.css" />
22
23    <!-- CSS Implementing Plugins -->
24    <link rel="stylesheet" href="assets/plugins/line-icons-pro/styles.css" />
25    <link rel="stylesheet" href="assets/plugins/line-icons/line-icons.css" />
26
27    <link
28      rel="stylesheet"
29      href="assets/plugins/font-awesome/css/font-awesome.min.css"
30    />
31
32    <!-- CSS Customization -->
33    <link rel="stylesheet" href="assets/css/custom.css" />
34  </head>
```

appointment page:

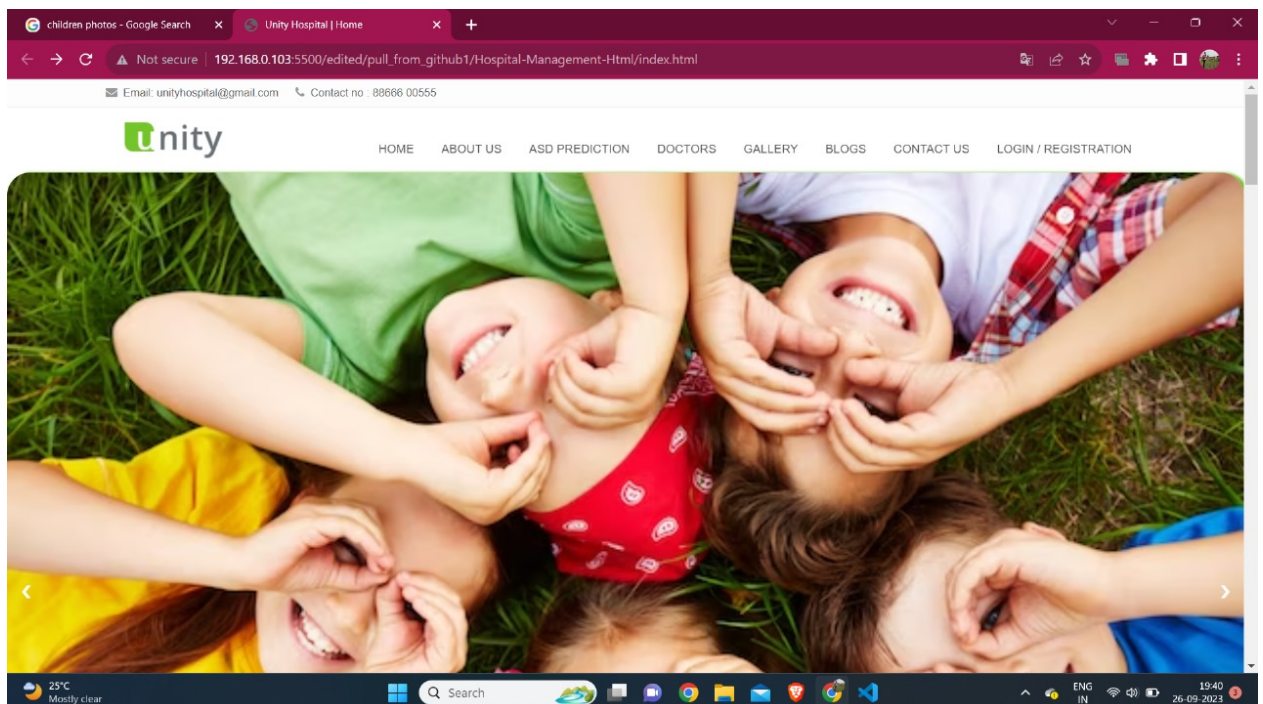
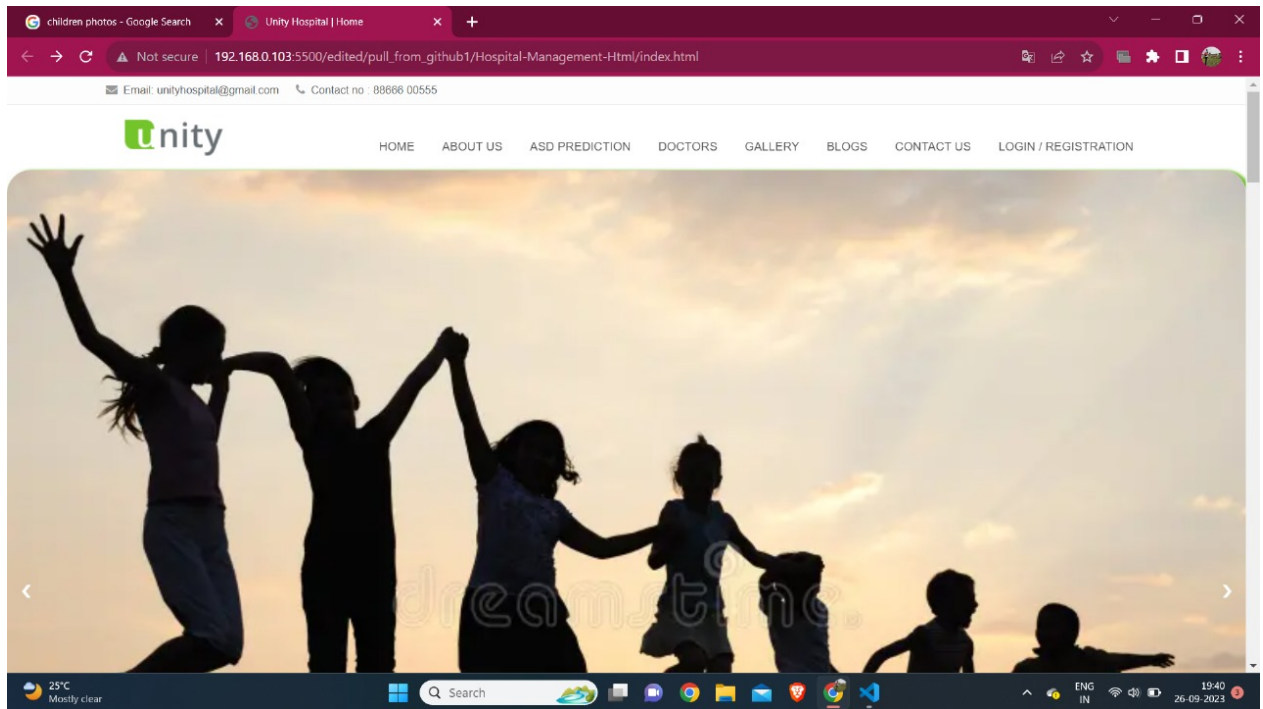


```
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blog.html M doctors.html M contact.html M loginreg.html U appointment.html M JS connectjs U X app.py U JS loginregjs U user.js U ...
edited > pull_from_github1 > Hospital-Management-HTML > JS connectjs
1 // book appointment
2
3 const firebaseConfig = {
4   apiKey: "AIzaSyC1l_opWA2cz3BSWm9CJX5uXrT0A64Ro9E",
5   authDomain: "mlproject-a5a5c.firebaseio.com",
6   databaseURL: "https://mlproject-a5a5c-default-rtdb.firebaseio.com",
7   projectId: "mlproject-a5a5c",
8   storageBucket: "mlproject-a5a5c.appspot.com",
9   messagingSenderId: "343935944581",
10  appId: "1:343935944581:web:4e2e160d62c90c6a31e6ec"
11 };
12 firebase.initializeApp(firebaseConfig);
13
14 var mlproj=firebase.database().ref('mlproject');
15 document.getElementById('appointmentForm').addEventListener('submit',submitForm);
16
17 function submitForm(e){
18   e.preventDefault();
19   var emailid=getElementByVal('email');
20   var name=getElementByVal('name');
21   var number=getElementByVal('number');
22   var date=getElementByVal('date');
23   var time=getElementByVal('time');
24   var doc=getElementByVal('doc');
25   savemsg(emailid,name,number,date,time,doc);
26
27 }
28
29
30 const savemsg=(emailid,name,number,date,time,doc)=>{
31   var newContactForm=mlproj.push();
32   newContactForm.set({
33     emailid:emailid,
34     name:name,
35     number:number,
36     date:date,
37     time:time,
38     doc:doc
39   });
40 }
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CHAPTER 7

RESULT

Home Page:



Form Page:

The screenshot shows a web browser window with the URL `localhost:8501`. The browser tabs include 'app - Streamlit', 'Login + Firebase Database', 'User Profile', and 'Unity Hospital | About Us'. The page title is 'Autism Prediction App'. The form contains the following questions and options:

Please answer the following questions:

- I often notice small sounds when others do not:
☐ Yes
☒ No
- I usually concentrate on whole picture rather than small details:
☐ Yes
☒ No
- I find it easy to do more than one thing at once:
☐ Yes
☒ No
- If there is an interruption i can switch back to what i was doing very quickly:
☐ Yes
☒ No

The Windows taskbar at the bottom shows the date and time as 11:47 on 26-09-2023.

The screenshot shows the result page of the 'Autism Prediction App'. The form contains the following information:

- ☐ No
- Any family member diagnosed with autism?
☒ Yes
☐ No
- Ethnicity
Hispanic
- Submit

Prediction Result:

Autism detected.

Please visit [this link](#) for more information.

Made with Streamlit

The Windows taskbar at the bottom shows the date and time as 10:51 on 26-09-2023.

Login/Register Page:

Email: unityhospital@gmail.com Contact no : 88666 00555

Unity

HOME ABOUT US DOCTORS GALLERY BLOGS CONTACT US

LOGIN + FIREBASE DATABASE

Full name

Email

New Password

Login Register

25°C Mostly sunny 11:15 26-09-2023

Appointment form:

Email: unityhospital@gmail.com Contact no : 88666 00555

Unity

HOME ABOUT US DOCTORS GALLERY BLOGS CONTACT US REGISTRATION LOGIN BOOK APPOINTMENT

Appointment Form

Name E-mail

Mobile Number

Choose available doctors:

Select Date Select Time

Submit

Appointment Notes

You Only Can Book Your Appointment Between 8 AM to 10 PM.
In Other Times You Can Call Our Ambulance Which Is Available 24/7.

Business Hours

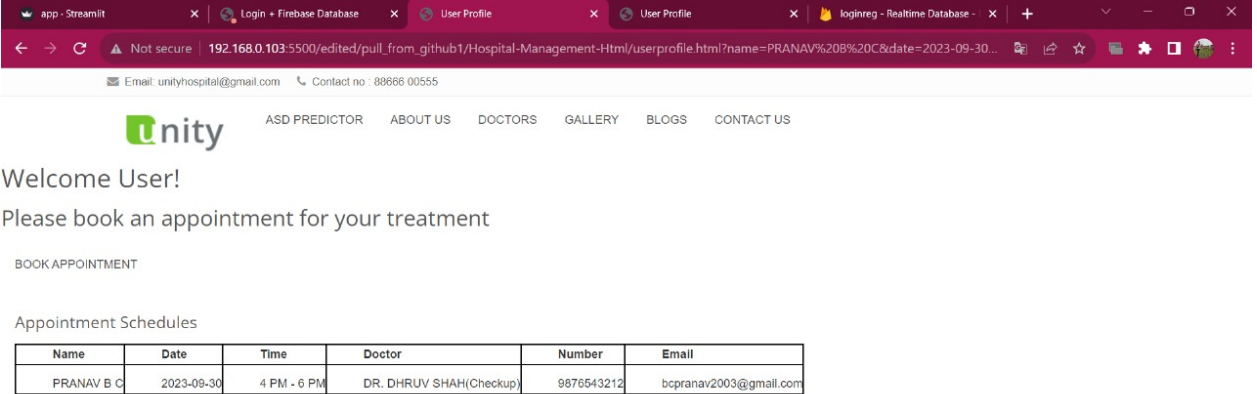
Monday-Saturday: 24/7 Available.
Sunday: 4 AM to 11 PM.

Why Choose Us?

All healthcare facilities can be accessed here under one roof, making UNITY Hospital a one point contact for all your healthcare.

25°C Near record 11:16 26-09-2023

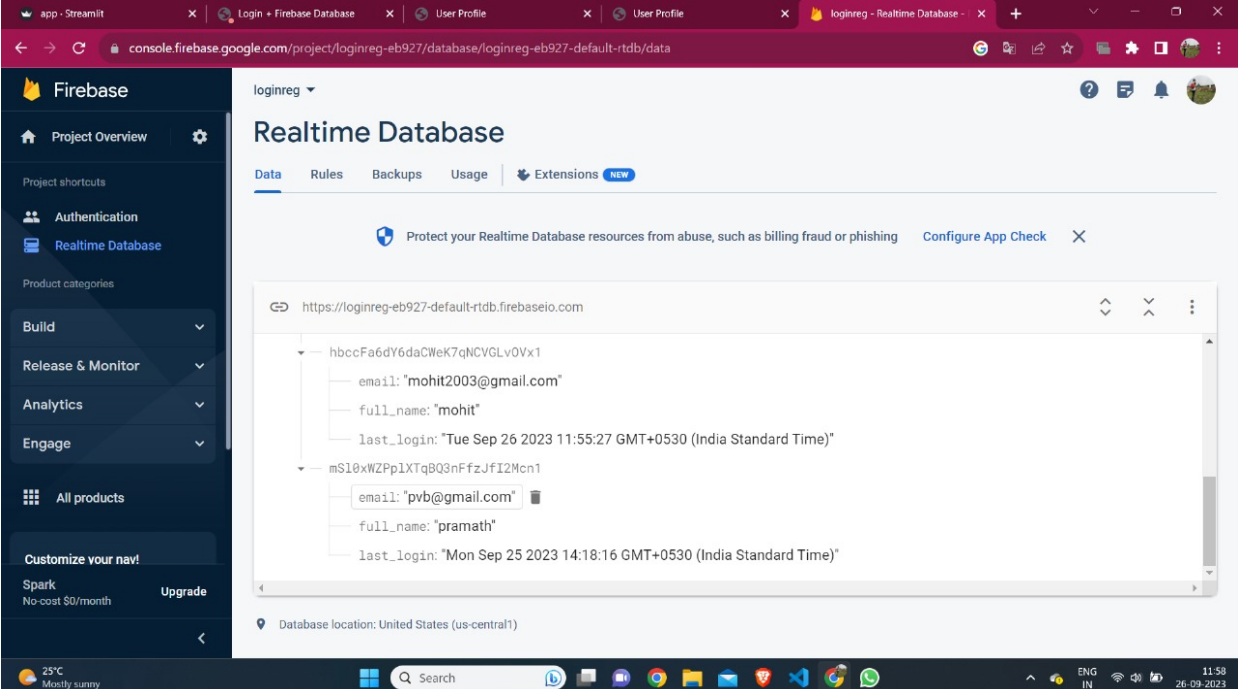
Appointment :



The screenshot shows a web browser with multiple tabs open. The active tab displays a user profile page for 'Unity'. The page includes a navigation bar with links: ASD PREDICTOR, ABOUT US, DOCTORS, GALLERY, BLOGS, and CONTACT US. Below the navigation bar, there is a welcome message and a prompt to book an appointment. A table titled 'Appointment Schedules' lists appointment details for a user named PRANAV B C.

Name	Date	Time	Doctor	Number	Email
PRANAV B C	2023-09-30	4 PM - 6 PM	DR. DHIRUV SHAH(Checkup)	9876543212	bcpranav2003@gmail.com

Firestore:



The screenshot shows the Firebase Realtime Database console. The left sidebar contains navigation options: Project Overview, Authentication, Realtime Database, Build, Release & Monitor, Analytics, Engage, All products, and Customize your nav!. The main area displays the Realtime Database structure for the 'loginreg' database. The data is organized into two nodes, each containing user information.

```

{
  hbccFa6dY6daCWeK7qNCVGLvOVx1: {
    email: "mohit2003@gmail.com"
    full_name: "mohit"
    last_login: "Tue Sep 26 2023 11:55:27 GMT+0530 (India Standard Time)"
  },
  mS10xWZPp1XTqBQ3nFfzJfI2Mc1: {
    email: "pvb@gmail.com"
    full_name: "pramath"
    last_login: "Mon Sep 25 2023 14:18:16 GMT+0530 (India Standard Time)"
  }
}
  
```

The database location is set to United States (us-central1).

CHAPTER 8

CONCLUSION

8.1 Conclusion

The development of the Autism Spectrum Disorder Detection System marks a significant step forward in the realm of healthcare and technology. Through the careful integration of machine learning algorithms, data analysis, and an extensive dataset, we have strived to create a tool that not only aids in the early diagnosis of Autism but also offers valuable insights for intervention and treatment strategies.

As we reflect on the journey of this project, we recognize the immense potential it holds in improving the lives of individuals and families affected by Autism. However, we also acknowledge the ongoing need for collaboration with healthcare professionals, ethical considerations, and continuous refinement to ensure its effectiveness and ethical use.

With a commitment to empathy, inclusivity, and rigorous research, we aspire to see this system contribute positively to the broader conversation surrounding Autism Spectrum Disorder diagnosis and care. Our hope is that this project will serve as a catalyst for further innovation and support in the field of Autism research and assistance.

8.2 FUTURE ENHANCEMENT

We can add additional features such as booking appointments in our web application, users can search for suitable doctors depending on their condition and schedule their visits online.

An additional feature can be added where the user can be directed to the nearest hospital or clinic with respect to their symptoms.

CHAPTER 9

REFERENCES

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